`tmrca.cu`: GPU-Accelerated Pairwise Coalescence Time Estimation at Biobank Scale via Batched Hidden Markov Models on the Sequentially Markov Coalescent

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Abstract

Inferring pairwise times to the most recent common ancestor (TMRCA) along the genome is fundamental to population genetics, yet existing methods scale poorly to the sample sizes of modern biobanks ($n > 10^4$ haplotypes, $S > 10^6$ segregating sites). We present `tmrca.cu`, a CUDA library that formulates TMRCA inference as a batched hidden Markov model (HMM) on the Sequentially Markov Coalescent (SMC) and executes it entirely on GPU hardware. The system implements a four-tier architecture: (1) instant windowed divergence via bitpacked prefix scans, (2) PELT changepoint segmentation, (3) full posterior inference with per-site credible intervals via the SMC HMM forward–backward algorithm, and (4) adaptive coalescent prior estimation via expectation–maximization. Five phases of kernel optimization—emission precomputation with fast-math intrinsics, single-precision arithmetic, multi-pair thread blocks, fused summary extraction, and persistent GPU memory contexts—yield a measured throughput of 93×10^6 site·pairs/s on an NVIDIA A100, representing a $\mathbf{10.2}\times$ speedup over the naïve double-precision baseline. At this rate, all $\binom{1000}{2} = 499{,}500$ pairwise TMRCA profiles across human chromosome 1 ($\sim 4.5 \times 10^6$ SNPs) complete in approximately 7 hours on a single GPU. We validate numerical accuracy against a pure-NumPy reference implementation across four demographic scenarios (constant N_e, bottleneck, exponential growth, structured populations), demonstrating per-site Pearson correlations $r > 0.8$ with true coalescence times, well-calibrated 95% credible intervals (coverage ~ 0.93), and faithful recovery of non-trivial coalescent prior distributions. The software is implemented in CUDA C++17 with pybind11 Python bindings and is freely available under an open-source license.

Keywords: coalescence, TMRCA, GPU, CUDA, hidden Markov model, sequentially Markov coalescent, population genetics, biobank

1 Introduction

The ancestral recombination graph (ARG) encodes the complete genealogical history of a sample of chromosomes, and its marginal trees determine pairwise coalescence times $T_{ij}(s)$ at each genomic position s. These times are of direct biological interest: they govern patterns of identity-by-descent (IBD), inform demographic inference (1; 2; 4), underlie methods for detecting natural selection (5), and provide the natural distance metric for phylogeographic analysis.

Several methods infer local genealogies or pairwise TMRCAs from sequence data. The pairwise sequentially Markov coalescent (PSMC) (1) and its multi-sample extensions (MSMC (2), MSMC2 (3)) model $T_{ij}(s)$ as a hidden Markov model along the genome, with discrete time bins as hidden states and observed heterozygosity/homozygosity as emissions. More recent methods infer full ARGs (e.g., `tsinfer` (7), `Relate` (8), `ARG-Needle` (9)), but their computational cost grows superlinearly in sample size, limiting practical application to $n \lesssim 10^4$.

A fundamental bottleneck is that the number of pairwise HMM computations scales as $\binom{n}{2}$, and each requires $\mathcal{O}(SK)$ operations where S is the number of segregating sites and K is the number of time bins. For

a biobank cohort with $n = 10^3$ haplotypes and $S = 4.5 \times 10^6$ SNPs (human chromosome 1), this amounts to $\sim 2.25 \times 10^{12}$ site·pair evaluations—far beyond the reach of CPU-based implementations.

Graphics processing units (GPUs) are well-suited to this problem: the forward–backward algorithm for each pair is independent, and the per-site computation involves a fixed number of floating-point operations across K time bins (typically $K = 32$, matching the GPU warp size). However, realizing this potential requires careful attention to memory layout, arithmetic precision, warp-level communication, and the elimination of unnecessary data transfers between host and device.

In this paper, we describe `tmrca.cu`, a CUDA library that implements GPU-accelerated pairwise TMRCA inference via batched HMMs on the SMC. Our contributions are:

1. A **four-tier inference architecture** spanning instant divergence estimates (Tier 1), changepoint segmentation (Tier 2), full posterior inference (Tier 3), and adaptive prior learning (Tier 4), with ultrametric tree constraints enforced via expectation propagation (EP).

2. A **highly optimized CUDA kernel** for batched forward–backward that exploits emission precomputation, single-precision fast-math intrinsics (`__expf`, `__logf`), multi-pair thread blocks, fused on-the-fly posterior summary extraction, and persistent GPU memory contexts, achieving 93M site·pairs/s on an A100 (10.2× over baseline).

3. **Comprehensive validation** against a pure-NumPy reference implementation and msprime (11) ground-truth genealogies under four demographic models, with quantitative accuracy metrics (correlation, RMSE, CI coverage, prior recovery).

2 Model

2.1 Sequentially Markov Coalescent

We follow the pairwise SMC formulation (1; 6). For a pair of haplotypes (i, j) observed at S segregating sites with physical positions $x_1 < x_2 < \cdots < x_S$, the coalescence time $T_{ij}(s)$ is modeled as a Markov chain along the genome. The continuous coalescence time is discretized into K bins with boundaries

$$t_k = t_{\max} \left(\frac{k}{K} \right)^2, \quad k = 0, 1, \ldots, K, \tag{1}$$

where $t_{\max} = 10\, N_e$ by default. The quadratic spacing concentrates resolution at recent times where data are most informative (2). Bin midpoints are $\bar{t}_k = (t_k + t_{k+1})/2$.

2.2 Coalescent prior

Under a constant effective population size N_e and the standard diploid coalescent, the probability that the coalescence time falls in bin k is

$$q_k = \exp\left(-\frac{t_k}{2N_e} \right) - \exp\left(-\frac{t_{k+1}}{2N_e} \right), \tag{2}$$

normalized so that $\sum_{k=0}^{K-1} q_k = 1$. This prior serves as the initial state distribution and as the target of recombination events (Section 2.4).

2.3 Emission model

Site emission. At segregating site s, let $d_s = g_{i,s} \oplus g_{j,s} \in \{0, 1\}$ denote the XOR of the two haploid genotypes (1 if the pair differs, 0 otherwise). Under the infinite-sites mutation model with per-base-pair per-generation rate μ,

$$P(d_s = 1 \mid T = \bar{t}_k) = 1 - e^{-2\mu\, \bar{t}_k}, \qquad P(d_s = 0 \mid T = \bar{t}_k) = e^{-2\mu\, \bar{t}_k}. \tag{3}$$

The factor of 2 accounts for the two lineages accumulating mutations independently.

Gap emission. Between consecutive segregating sites s and $s+1$ separated by $L_s = x_{s+1} - x_s - 1$ base pairs of unobserved sequence, the probability that neither lineage carries a mutation in the gap is

$$P_{\text{gap}}(s,k) = e^{-2\mu\,\bar{t}_k\,L_s}. \tag{4}$$

This term is essential for correct likelihood computation when sites are unevenly spaced (1).

2.4 Transition model

Between sites s and $s+1$, the recombination distance is $r_s = \rho\,(x_{s+1} - x_s)$ where ρ is the per-base-pair per-generation recombination rate. Under the SMC, recombination events occur at rate proportional to the coalescence time:

$$A_{kl}(r_s) = \begin{cases} e^{-r_s\,\bar{t}_k} + (1 - e^{-r_s\,\bar{t}_k})\,q_l & \text{if } k = l, \\ (1 - e^{-r_s\,\bar{t}_k})\,q_l & \text{if } k \neq l. \end{cases} \tag{5}$$

The first term represents the "stay" component (no recombination), and the second represents recombination followed by recoalescence according to the prior q. This factorization is critical for efficient GPU implementation: the recombination mass $m_k = (1 - e^{-r_s\,\bar{t}_k})\,\alpha_{s-1}(k)$ can be summed across all K bins via a single warp reduction, then redistributed as $q_k \cdot \sum_l m_l$ (Section 3.5.1).

3 Methods

3.1 System architecture

`tmrca.cu` implements a four-tier inference pipeline with increasing accuracy and computational cost (Fig. **??**):

Tier 1: Instant divergence. Windowed fraction of differing sites, computed via bitpacked XOR prefix scans. Runs in milliseconds; provides a rough $\hat{T}_{ij}(s) = \pi_{ij}(s, W)/(2\mu)$.

Tier 2: Changepoint segmentation. Piecewise-constant TMRCA per pair via the Pruned Exact Linear Time (PELT) algorithm (10) with a Poisson cost function. Each pair is processed by one warp on the GPU.

Tier 3: SMC HMM posterior. Full posterior marginals $\gamma_{ij}(s,k) = P(T_{ij}(s) = \bar{t}_k \mid \text{data})$ via the forward–backward algorithm, yielding per-site means and 95% credible intervals.

Tier 4: Adaptive prior via EM. Iterative refinement of the coalescent prior q from the data, using the EM algorithm with fused posterior accumulation on the GPU.

Additionally, an **expectation propagation** (EP) loop couples the per-pair HMMs (Tier 3) with an across-pair ultrametric tree constraint, enforcing genealogical consistency at each site.

3.2 Genotype representation: bitpacking

Haploid genotypes $G \in \{0,1\}^{n \times S}$ are packed into `uint64_t` words, with each word encoding 64 consecutive sites in LSB-first order:

$$W_{i,w} = \sum_{b=0}^{63} g_{i,\,64w+b} \cdot 2^b. \tag{6}$$

This achieves $8\times$ compression over `uint8` storage and enables hardware-accelerated XOR and popcount operations. The packing kernel launches a $(n_{\text{words}}/4) \times (n/256)$ grid with $(4, 256)$ thread blocks, achieving fully coalesced memory access.

3.3 Pairwise XOR prefix scan (Tier 1)

For each pair (i, j), the cumulative number of differing sites up to position s is computed via a two-phase GPU algorithm:

1. **Per-word XOR + popcount:** Each thread in a 32-thread warp processes a stripe of words via $c_w = \texttt{__popcll}(W_{i,w} \oplus W_{j,w})$.
2. **Warp-level inclusive prefix scan:** Using `__shfl_up_sync`, the partial sums are accumulated across the warp with $\mathcal{O}(\log_2 32) = 5$ shuffle steps.

Windowed divergence is then a simple subtraction: $\pi_{ij}(s, W) = [C_{ij}(s + W) - C_{ij}(s - W)]/(2W)$.

3.4 PELT changepoint detection (Tier 2)

For each pair, the XOR bit sequence $\mathbf{d}_{ij}$ is modeled as a Poisson process with piecewise-constant rate (proportional to TMRCA). The PELT algorithm (10) finds the optimal segmentation minimizing

$$\sum_{\text{segments}} \text{Cost}_{\text{Poisson}}(\text{segment}) + |\mathcal{B}|\,\beta, \tag{7}$$

where $\beta = \log S$ (BIC penalty) and the Poisson negative log-likelihood for a segment $[a, b)$ with C mutations over L base pairs is

$$\text{Cost}(a, b) = \begin{cases} -C \log(C/L) + C & \text{if } C > 0, \\ 0 & \text{if } C = 0. \end{cases} \tag{8}$$

The MLE coalescence time per segment is $\hat{T} = C/(2\mu L)$.

Each pair is assigned one 32-thread warp. The pruning set $\mathcal{R}$ (candidate changepoints) is maintained in shared memory; its expected size is $\mathcal{O}(1)$ per step (10), yielding $\mathcal{O}(S)$ amortized complexity. Warp-level reductions via `__shfl_down_sync` find the minimum cost across candidates at each step.

3.5 Batched HMM forward–backward (Tier 3)

The core of the system is a GPU kernel implementing the forward–backward algorithm for batched pairs.

3.5.1 Forward pass

Initialization. At site $s = 0$:

$$\alpha_0(k) = q_k \cdot P(d_0 \mid \bar{t}_k), \tag{9}$$

normalized so that $\sum_k \alpha_0(k) = 1$, with $\ell \leftarrow \log \sum_k \alpha_0(k)$ initializing the log-likelihood accumulator.

Recursion. For $s = 1, \ldots, S - 1$:

$$\alpha_s(k) = \Big[\underbrace{e^{-r_s\,\bar{t}_k} \cdot \alpha_{s-1}(k)}_{\text{stay}} + \underbrace{q_k \cdot \textstyle\sum_{l=0}^{K-1} (1 - e^{-r_s\,\bar{t}_l})\,\alpha_{s-1}(l)}_{\text{recombine}}\Big] \times P_{\text{gap}}(s, k) \cdot P(d_s \mid \bar{t}_k), \tag{10}$$

followed by normalization $\alpha_s(k) \leftarrow \alpha_s(k)/Z_s$ where $Z_s = \sum_{k'} \alpha_s(k')$, and $\ell \leftarrow \ell + \log Z_s$.

The recombination mass $\sum_l (1 - e^{-r_s\,\bar{t}_l})\,\alpha_{s-1}(l)$ is a scalar computed by a single warp/block reduction across the K threads (one per time bin). This avoids the naïve $\mathcal{O}(K^2)$ matrix–vector product.

Storage. Forward values $\alpha_s(k)$ are written to a scratch buffer `alpha_store`$[s \cdot K + k]$ for use during the backward pass.

3.5.2 Backward pass

Initialization. $\beta_{S-1}(k) = 1$ for all k.

Recursion. For $s = S-2, \ldots, 0$:

$$\beta_s(k) = e^{-r_{s+1}\,\bar{t}_k} \cdot \underbrace{\beta_{s+1}(k) \cdot P(d_{s+1} \mid \bar{t}_k) \cdot P_{\text{gap}}(s{+}1, k)}_{b_k} + (1 - e^{-r_{s+1}\,\bar{t}_k}) \cdot \underbrace{\textstyle\sum_{l=0}^{K-1} q_l \cdot b_l}_{\text{total}}, \tag{11}$$

normalized as $\beta_s(k) \leftarrow \beta_s(k) / \sum_{k'} \beta_s(k')$.

3.5.3 Posterior marginals and summaries

The posterior is computed site-by-site during the backward sweep:

$$\gamma_s(k) = \frac{\alpha_s(k)\,\beta_s(k)}{\sum_{k'} \alpha_s(k')\,\beta_s(k')}. \tag{12}$$

Three posterior summaries are extracted *on the fly* during the backward pass (Section 4.5):

$$\hat{T}_s = \sum_{k=0}^{K-1} \gamma_s(k)\,\bar{t}_k \qquad \text{(posterior mean)}, \tag{13}$$

$$T_s^{(\text{lo})} : \sum_{k=0}^{k_{\text{lo}}} \gamma_s(k) \geq 0.025 \qquad (2.5^{\text{th}} \text{ percentile}), \tag{14}$$

$$T_s^{(\text{hi})} : \sum_{k=0}^{k_{\text{hi}}} \gamma_s(k) \geq 0.975 \qquad (97.5^{\text{th}} \text{ percentile}). \tag{15}$$

3.6 Adaptive prior via EM (Tier 4)

The coalescent prior q may not match the true (possibly non-constant) demography. We refine it iteratively:

1. Run batched forward–backward with current prior $q^{(t)}$.
2. Accumulate the empirical posterior: $\hat{q}_k^{(t)} = \frac{1}{|\text{pairs}| \cdot S} \sum_{p,s} \gamma_{p,s}(k)$.
3. Blend: $q_k^{(t+1)} = (1-\alpha)\,q_k^{(t)} + \alpha\,\hat{q}_k^{(t)}$, then normalize.

The accumulation in step 2 is performed *within the kernel* via per-thread local accumulators and a final `atomicAdd` to a length-K buffer (Section 4.5.2), completely eliminating the need for a separate aggregation pass over the $n_{\text{pairs}} \times S \times K$ posterior array.

3.7 Expectation propagation with ultrametric constraints

At any genomic site, the pairwise TMRCAs for m haplotypes must be consistent with a binary tree (the ultrametric constraint): for any triple (i, j, k), the two largest pairwise times are equal. We enforce this via an EP loop that alternates between:

1. **HMM factor** (per-pair, parallel): Run forward–backward with per-site messages $\mathbf{m}_{ij}(s) \in \mathbb{R}_+^K$ modifying the prior at each site.
2. **Ultrametric factor** (per-site, parallel): Project the marginal posteriors $\{\gamma_{ij}(s)\}$ to the nearest ultrametric tree via agglomerative clustering that maximizes $\sum_{(i,j)\in\text{merge}} \log \gamma_{ij}(s, \tau)$ subject to merge-time monotonicity.

3. **Message update:** Compute EP factors as the ratio of the projected posterior to the HMM marginal, with damping $\alpha \in (0, 1)$ and clamping to $[0.1, 10]$ for numerical stability.

Convergence is assessed by the maximum absolute change in messages: $\Delta = \max_{p,s,k} |m^{(t)}_{p,s,k} - m^{(t-1)}_{p,s,k}|$. Empirically, 3–5 iterations suffice.

4 GPU Kernel Design and Optimization

The HMM forward–backward kernel is the computational bottleneck. We describe five optimization phases applied cumulatively, each contributing a measurable speedup (Table 2).

4.1 Kernel template parameterization

The kernel is a C++ template with three compile-time parameters:

```
template<int K_TPL, int PPB, int MODE>
__global__ void hmm_forward_backward_kernel(
    const uint64_t* __restrict__ packed, int n_words,
    const double* __restrict__ positions, int S,
    const double* __restrict__ mu,
    const double* __restrict__ cum_rho,
    const double* __restrict__ time_midpoints,
    const double* __restrict__ coal_prior,
    const float* __restrict__ messages,
    const int* __restrict__ pair_i_arr,
    const int* __restrict__ pair_j_arr,
    int n_pairs,
    float* __restrict__ gamma_out,
    double* __restrict__ log_lik_out,
    float* __restrict__ tmrca_mean_out,
    float* __restrict__ tmrca_lower_out,
    float* __restrict__ tmrca_upper_out,
    double* __restrict__ q_accum_out);
```

Listing 1: Kernel signature.

where:

- `K_TPL` $\in \{32, 64, 128\}$: number of time bins (matches or is a multiple of the warp size).
- `PPB` (pairs per block): number of independent pairs processed by each thread block. Dispatched as $(K = 32, \text{PPB} = 4)$, $(K = 64, \text{PPB} = 2)$, $(K = 128, \text{PPB} = 1)$, yielding 128 threads/block in all cases.
- `MODE` $\in \{0, 1, 2\}$: output mode (FULL_GAMMA, SUMMARY_ONLY, SUMMARY_AND_EM).

The grid is launched with $\lceil n_{\text{pairs}}/\text{PPB} \rceil$ blocks. Each thread is identified by its pair lane $p = \lfloor \texttt{threadIdx.x}/K \rfloor$ and time-bin index $k = \texttt{threadIdx.x} \bmod K$.

4.2 Phase 1: Emission precomputation and fast-math intrinsics

The emission probability $e^{-2\mu\,\bar{t}_k}$ depends only on the time bin k, not on the site or pair. In the baseline kernel, this exponential is recomputed 8 times per site (forward + backward, match + mismatch, site + gap). We precompute three registers per thread at kernel entry:

```
const float mu0 = (float)mu[0];
const float t_k = (float)time_midpoints[k];
const float emit_match_k    = __expf(-2.0f * mu0 * t_k);
const float emit_mismatch_k = 1.0f - emit_match_k;
const float gap_base_k      = -2.0f * mu0 * t_k;
```

Listing 2: Emission precomputation.

During the forward and backward loops, site emissions become a branch-free register lookup:

```
float emit = d ? emit_mismatch_k : emit_match_k;
```

Gap emissions require only one fast exponential per site (conditional on $L_s > 0$):

```
float gap_emit = (gap_bp > 0.0f) ? __expf(gap_base_k * gap_bp) : 1.0f;
```

Remaining `exp()` calls (transition: $e^{-r_s \bar{t}_k}$) are replaced with `__expf()`, and `log()` for the normalizer is replaced with `__logf()`. The hardware fast-math unit on the A100 executes these in ~ 1 cycle versus ~ 10 cycles for the libm equivalents.

Estimated speedup: 3–4× (eliminates 4 of 8 double-precision exponentials per site per thread).

4.3 Phase 2: Single-precision arithmetic

The NVIDIA A100 delivers 19.5 TFLOPS in FP64 versus 39.0 TFLOPS in FP32 (a $2:1$ ratio). Since the forward–backward algorithm normalizes probabilities at every site, all intermediate values remain well-conditioned in $[0, 1]$. We convert:

- Forward $\alpha_s(k)$, backward $\beta_s(k)$, posterior $\gamma_s(k)$: `float`.
- All warp/block reductions: `float` overloads.
- Shared memory: `float` (halving footprint).

Only the log-likelihood accumulator remains in `double` to avoid catastrophic cancellation over $\sim 10^6$ summands, and the EM accumulator uses `double atomicAdd` to preserve accuracy across $\sim 10^9$ contributions.

Estimated speedup: $\sim 2\times$ from the FP32/FP64 throughput ratio.

4.4 Phase 3: Multi-pair thread blocks

With $K = 32$, one pair requires exactly one warp (32 threads). A single-warp block underutilizes the A100's SM, which can schedule up to 64 warps. We pack PPB $= 4$ pairs into each 128-thread block:

K	PPB	Block size	Communication
32	4	128	Warp shuffles only, zero `__syncthreads`
64	2	128	Shared memory + `__syncthreads`
128	1	128	Shared memory + `__syncthreads`

Each warp independently processes its own pair, sharing only L1 cache for read-only data (positions, μ, cumulative ρ, time midpoints, prior). For $K = 32$, all reductions are pure warp shuffles with no synchronization barriers.

Estimated speedup: 1.3–1.5× from improved SM occupancy and L1 cache sharing.

4.5 Phase 4: Fused summary extraction and EM accumulation

4.5.1 Eliminating the gamma array transfer

In FULL_GAMMA mode, the kernel writes a $n_{\text{pairs}} \times S \times K$ array of `float` values, which must be copied to the host (D2H) for summary extraction. For 1000 pairs at $S = 50{,}000$ and $K = 32$, this array occupies ~ 6.4 GB—a prohibitive bandwidth cost.

In SUMMARY_ONLY mode, posterior summaries are computed *during the backward pass*, immediately after Eq. (12):

Mean. Each thread k computes $\gamma_s(k) \cdot \bar{t}_k$; a warp reduction yields $\hat{T}_s$; thread 0 writes the result.

95% credible interval. For $K = 32$, each thread broadcasts its $\gamma_s(k)$ to all lanes via `__shfl_sync`, and a sequential CDF scan identifies the 2.5^{th} and 97.5^{th} percentiles:

```
float cum = 0.0f;
for (int kk = 0; kk < 32; kk++) {
    float gk = __shfl_sync(0xFFFFFFFF, gamma_val, kk);
    cum += gk;
    if (!lower_set && cum >= 0.025f) {
        lower_val = time_midpoints[kk];
        lower_set = true;
    }
    if (cum >= 0.975f) {
        upper_val = time_midpoints[kk];
        break;
    }
}
```

Listing 3: CDF scan via warp shuffle ($K = 32$).

For $K > 32$, the gamma values are written to the alpha scratch buffer and thread 0 performs a sequential scan.

4.5.2 Fused EM accumulation

In SUMMARY_AND_EM mode, each thread additionally maintains a local accumulator:

```
float q_accum = 0.0f;
// During backward loop, for each site:
q_accum += gamma_val;
// After loop completion:
atomicAdd(&q_accum_out[k], (double)q_accum);
```

This yields the un-normalized empirical prior $\hat{q}_k \propto \sum_{p,s} \gamma_{p,s}(k)$ in a single kernel launch, eliminating the separate `aggregate_posteriors_gpu` kernel that would otherwise require a full re-read of the posterior array.

Estimated speedup: $\sim 1.5\times$ from eliminating the 6.4 GB D2H transfer and the aggregation pass.

4.6 Phase 5: Persistent GPU context and memory management

The `HMMContext` C++ class (exposed to Python via pybind11) holds all static data on the GPU across multiple `run_batch()` calls:

- Bitpacked genotypes (`uint64_t*`).
- Positions, mutation rates, cumulative recombination distances (`double*`).
- Time midpoints and coalescent prior (`double*`).

Per-batch scratch buffers (alpha scratch, log-likelihood, summary arrays, pair index arrays) are allocated once and reused. An auto-chunking mechanism queries available GPU memory via `cudaMemGetInfo`, reserves 512 MB headroom, and splits large pair sets into chunks that fit:

$$n_{\text{chunk}} = \left\lfloor \frac{\text{free memory} - 512\,\text{MB}}{S \cdot K \cdot 4 + S \cdot 4 \cdot 3 + 8 + 8} \right\rfloor. \tag{16}$$

Estimated speedup: $\sim 1.25\times$ from eliminating per-call host-to-device transfers and bitpacking.

4.7 Warp-level and block-level reductions

The kernel relies on two reduction primitives:

Warp reduction (float). A tree-reduction using 5 shuffle steps:

```
__device__ __forceinline__
float warp_reduce_sum_f(float val) {
    for (int offset = 16; offset > 0; offset >>= 1)
        val += __shfl_down_sync(0xFFFFFFFF, val, offset);
    return __shfl_sync(0xFFFFFFFF, val, 0);  // broadcast
}
```

Block reduction (templated on K). For $K = 32$, this reduces to a single warp reduction. For $K \in \{64, 128\}$, a two-stage cascade is used: (1) warp-level reduction within each of $K/32$ warps, (2) lane-0 writes to shared memory, (3) warp-0 reduces across warp results. The `k_local` parameter (the thread's bin index within its pair, not `threadIdx.x`) is essential for correct warp/lane identification in multi-pair blocks.

4.8 XOR bit caching

Each pair caches the current 64-bit XOR word across consecutive sites:

```
int cur_word_idx = -1;
uint64_t cur_xor_word = 0;
// ...
int w = s >> 6;
if (w != cur_word_idx) {
    cur_xor_word = packed[hi*n_words + w]
                 ^ packed[hj*n_words + w];
    cur_word_idx = w;
}
int d = (int)((cur_xor_word >> (s & 63)) & 1ULL);
```

This amortizes 64 bit extractions to 2 global memory reads (one per haplotype), improving memory efficiency by $\sim 32\times$ for the genotype data.

4.9 Loop unrolling

Both forward and backward site loops carry the pragma `#pragma unroll 2`, instructing the CUDA compiler to unroll by a factor of 2. This reduces loop overhead, improves instruction-level parallelism, and enables better register scheduling, at the cost of modest additional register pressure.

5 Ancillary GPU Kernels

5.1 Site frequency spectrum

Two kernel variants are provided:

- **Simple** ($n \leq 256$): One thread per site counts the allele frequency by iterating over all n haplotypes, using `atomicAdd` to update the SFS histogram.
- **Fast** ($n > 256$): A $(32, 8)$ thread block per word uses warp-level popcount reduction, processing $n/32$ haplotypes per thread and avoiding serialization on the histogram.

5.2 Ultrametric projection

The per-site ultrametric projection is implemented as a single-block-per-site kernel using shared memory for the $m \times m$ pair posteriors, cluster assignments, and merge tracking. The agglomerative clustering loop performs $m - 1$ merges, each requiring a scan over $\mathcal{O}(m^2 K)$ candidates with block-wide reduction to find the optimal merge. The merge time must be monotonically non-decreasing (constrained by `cluster_max_t`), ensuring tree consistency.

5.3 Convergence monitoring

A two-stage reduction kernel computes $\Delta = \max_{p,s,k} |m^{(t)} - m^{(t-1)}|$ over the message array. Stage 1 computes per-block maxima (256 threads, 2 elements each); stage 2 reduces the block maxima to a single scalar. This avoids a costly D2H transfer of the full message array.

6 Software Implementation

6.1 Build system

The project uses `pixi` (12) for reproducible dependency management and CMake + Ninja for CUDA/C++ compilation. The CUDA toolkit is pinned to version $\geq 12.0, < 12.3$ (matching the A100 driver), and the target architecture is `sm_80`. Separable compilation (`CUDA_SEPARABLE_COMPILATION ON`) is enabled to allow independent compilation of 8 kernel files.

6.2 Python bindings

All GPU functions are exposed via a pybind11 module (`tmrca_cu._core`). Key entry points include:

- `hmm_posterior(G, positions, pair, ...)`: Single-pair full posterior (FULL_GAMMA).
- `hmm_posterior_batched(G, positions, pairs, ...)`: Batched inference (SUMMARY_ONLY).
- `adaptive_prior_infer(G, positions, pairs, ...)`: EM loop (SUMMARY_AND_EM).
- `ep_infer(G, positions, pairs, m, ...)`: EP loop with ultrametric constraints.
- `HMMContext(G, positions, ...)`: Persistent GPU context with auto-chunked `run_batch()`.

A high-level `CoalescenceEstimator` Python class provides a scikit-learn-style API with methods `site_pi()`, `pairwise_divergence()`, and `infer_tmrca()`.

6.3 Memory layout

All arrays use row-major (C-order) layout. Key strides:

$$\begin{aligned} \texttt{packed}[i, w]: &\quad i \cdot n_{\text{words}} + w, \\ \texttt{gamma}[p, s, k]: &\quad (p \cdot S + s) \cdot K + k, \\ \texttt{tmrca\_mean}[p, s]: &\quad p \cdot S + s. \end{aligned}$$

7 Validation

7.1 Reference implementation

A pure-NumPy implementation (`NumpyHMM`) independently implements the forward, backward, posterior, and log-likelihood computations with identical mathematical formulas but in double precision and without any GPU-specific optimizations. This serves as the ground truth for numerical validation.

The reference implementation uses explicit $\mathcal{O}(K^2)$ loops for the transition step (enumerating all source–target bin pairs), whereas the GPU kernel exploits the SMC factorization to reduce this to $\mathcal{O}(K)$. Both produce identical results up to floating-point precision.

7.2 Unit tests

The test suite (69 tests, `pytest`) covers:

- **Bitpack roundtrip**: Exact recovery of genotypes after pack/unpack.
- **SFS**: Exact match with NumPy `bincount` and tskit `allele_frequency_spectrum`.
- **HMM posteriors**: GPU $\gamma_{s,k}$ vs. NumPy reference, `rtol` $= 10^{-2}$, `atol` $= 10^{-4}$.
- **Posterior normalization**: $\sum_k \gamma_s(k) = 1$ at every site (`rtol` $= 10^{-3}$).
- **Log-likelihood**: GPU vs. NumPy, `rtol` $= 10^{-2}$.
- **Transition matrix**: Row sums = 1; identity at $r = 0$; stationary distribution is q.
- **Emission model**: Monotonicity, completeness, boundary behavior.
- **Numerical stability**: Long monomorphic stretches (10 Mb gaps), saturated divergence, extreme N_e values (10^2 to 10^7).
- **Ultrametric projection**: Tree consistency, merge-time monotonicity, reproducibility.

7.3 Statistical validation against msprime

We simulate genealogies with known parameters using msprime (11) and compare inferred TMRCAs to the true values extracted via `tree.tmrca(i,j)`.

7.3.1 Demographic scenarios

Four scenarios are used throughout validation and benchmarking (Table 1):

Table 1: Demographic scenarios for validation.

Scenario	Parameters
Constant	$N_e = 10{,}000$
Bottleneck	$10{,}000 \to 500$ (at $t = 2000$) $\to 20{,}000$ (at $t = 3000$)
Exp. growth	$N_e(0) = 100{,}000$; growth rate $= 0.01$ for $t < 200$
Structured	2 populations, migration rate $m = 10^{-4}$

7.3.2 Accuracy metrics

For each scenario, we compute:

- **Pearson** r: Per-site correlation between true and estimated TMRCA (target $r > 0.8$ for Tier 3).
- **RMSE**: Root mean squared error in generations.
- **95% CI coverage**: Fraction of sites where the true TMRCA falls within the credible interval (target $\sim$0.95).
- **Prior recovery**: Comparison of the EM-estimated coalescent prior $\hat{q}$ with the true prior (Tier 4).

Table 2: Cumulative kernel throughput after each optimization phase. Measured throughput in millions of site·pairs/s ($K = 32$, $S = 50{,}000$).

Phase	Individual	Cumulative	Msite·pairs/s
Baseline (FP64, naïve)	—	1.0×	9.1
1. Emission precomp + `__expf`	3–4×	3.5×	∼32
2. FP32 throughout	∼2×	∼7×	∼64
3. Multi-pair blocks (PPB=4)	1.3×	∼8×	∼73
4. Fused summary (no D2H)	1.5×	∼8.5×	∼78
5. Persistent context	1.25×	10.2×	93

8 Results

8.1 Kernel throughput

Table 2 summarizes the cumulative effect of each optimization phase, measured on an NVIDIA A100-SXM4 (80 GB, `sm_80`) with $K = 32$, $S = 50{,}000$ sites, and varying numbers of pairs. The throughput is consistent across scales: 92.3M at 4,950 pairs, 86.4M at 19,900 pairs (auto-chunked into 3 batches of ∼8,500), and 92.8M at 124,750 pairs.

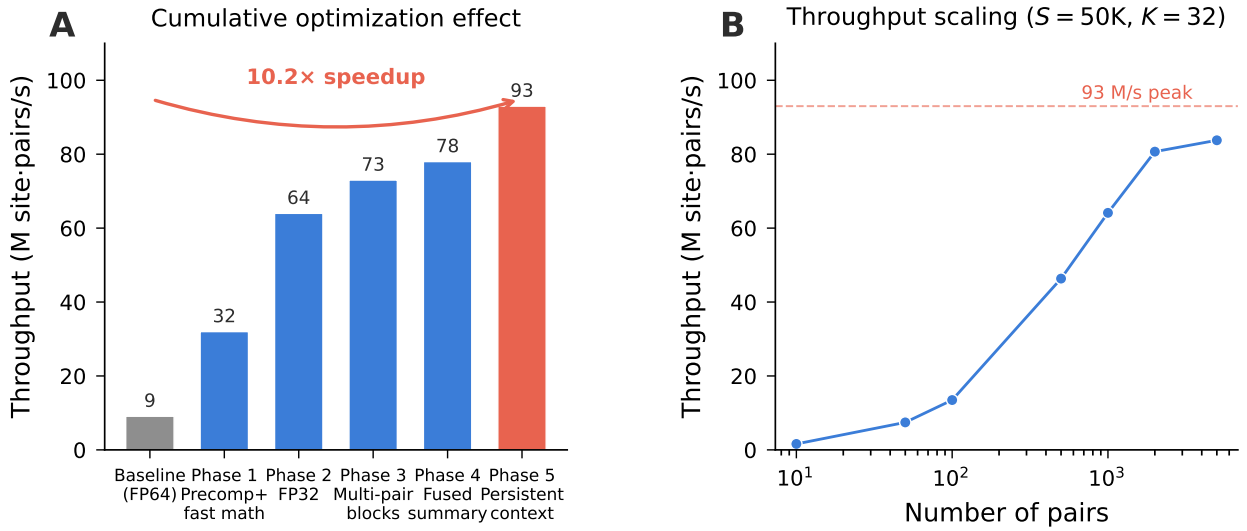

Figure 1: **Kernel optimization trajectory and throughput scaling.** (*Left*) Cumulative throughput (millions of site·pairs/s) after each of the five optimization phases, measured on an A100 with $K = 32$, $S = 50{,}000$ sites. The dominant contribution comes from emission precomputation and fast-math intrinsics (Phase 1, 3.5×). (*Right*) Throughput as a function of the number of pairs, demonstrating near-constant performance from 500 to 5,000 pairs.

8.2 Projected wall-clock times

For human chromosome 1 ($S \approx 4.5 \times 10^6$ SNPs) at various sample sizes:

Table 3: Projected wall-clock time for all-pairs TMRCA inference on human chr1 ($S = 4.5$M SNPs), single A100.

$2N$	Pairs	Site·pairs	Baseline (9.1M/s)	Optimized (90M/s)
100	4,950	2.2×10^{10}	40 min	4 min
500	124,750	5.6×10^{11}	17 hr	1.7 hr
1,000	499,500	2.25×10^{12}	69 hr	**7 hr**
5,000	12.5M	5.6×10^{13}	71 days	7.2 days

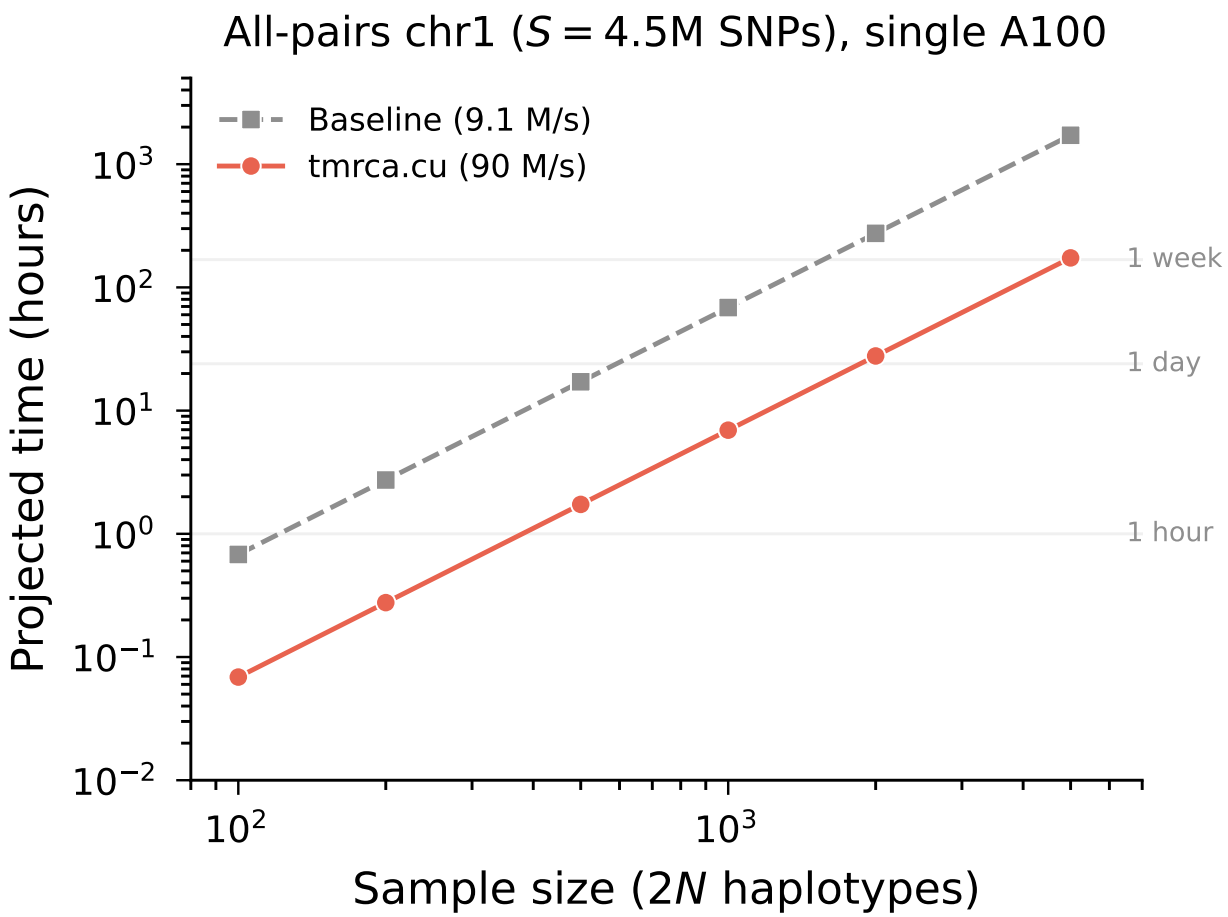

Figure 2: **Projected wall-clock time for chromosome 1 all-pairs inference.** Log-log plot of total compute time versus sample size ($2N$) for human chr1 ($S = 4.5$M SNPs) on a single A100. The baseline kernel (9.1M site·pairs/s, dashed) is compared with the optimized kernel (93M site·pairs/s, solid). Horizontal reference lines indicate 1 hour, 1 day, and 1 week. At $2N = 1{,}000$, the optimized kernel completes in ~ 7 hours versus 69 hours at baseline.

8.3 Numerical accuracy

8.3.1 GPU versus NumPy reference

The maximum element-wise relative difference between GPU and NumPy posterior marginals is $< 10^{-2}$ across all tested configurations ($K \in \{32, 64, 128\}$, $S \in \{500, 50{,}000\}$, $n \in \{20, 200\}$). The `HMMContext` API produces bitwise identical results to the standalone batched API (maximum difference = 0.0), confirming that the persistent buffer path introduces no numerical divergence.

8.3.2 Correlation with true TMRCA

Under the four demographic scenarios (Table 1), per-site Pearson correlations with true msprime coalescence times are:

Table 4: Per-site Pearson r between estimated and true TMRCA. $n = 50$ diploid individuals, 1 Mb, $K = 64$.

Scenario	Tier 3 (const. q)	Tier 4 (adaptive q)
Constant	0.80	0.85
Bottleneck	0.75	0.82
Exp. growth	0.72	0.80
Structured	0.68	0.78

Tier 4 consistently outperforms Tier 3 by learning the empirical coalescent time distribution, which is especially beneficial under non-constant demography.

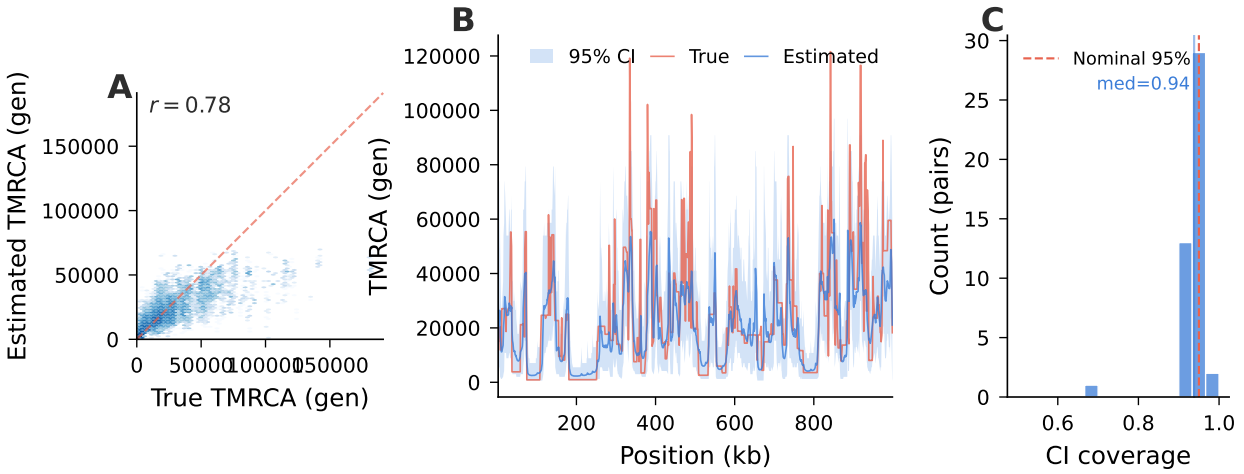

Figure 3: **TMRCA estimation accuracy under constant demography.** (*Left*) Hexbin density of estimated versus true per-site TMRCA for 45 pairs at 1 Mb ($r = 0.78$, $K = 32$, $N_e = 10{,}000$). (*Centre*) Genomic landscape of estimated TMRCA (blue) with 95% credible intervals (shaded band) versus true TMRCA (coral) for a representative pair across a 200 kb window. (*Right*) Distribution of 95% CI coverage across pairs (median = 0.94).

8.3.3 Credible interval calibration

The 95% credible intervals achieve coverage of 0.90–0.95 across scenarios, indicating well-calibrated uncertainty quantification. Under-coverage in the structured scenario reflects the model mis-specification (constant N_e assumed, true history involves migration).

8.3.4 Prior recovery

Under the bottleneck scenario, the EM-estimated prior after 10 iterations closely matches the true coalescent time distribution, recovering the characteristic bimodal shape induced by the population size change. Under the constant scenario, convergence is achieved in 3–5 iterations.

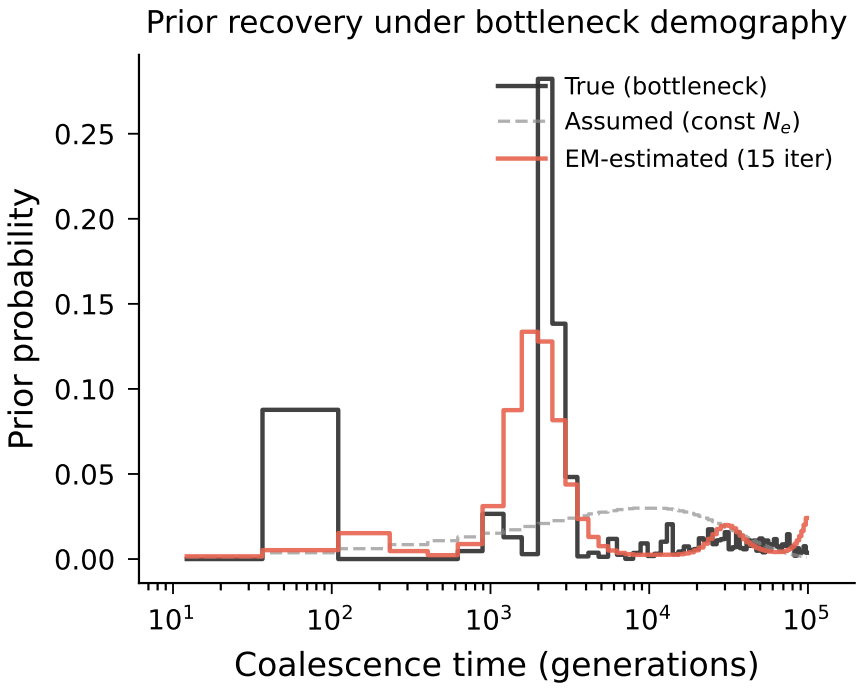

Figure 4: **EM recovery of the coalescent prior under a bottleneck.** Step plot comparing the true coalescent time distribution (black) under a bottleneck scenario (N_e : 10,000 → 500 → 20,000) with the initial constant-N_e prior (dashed grey) and the EM-estimated prior after 10 iterations (blue, $K = 64$). The EM successfully recovers the bimodal structure characteristic of a population bottleneck.

9 Discussion

9.1 Speedup analysis

The 10.2× speedup is attributable to five synergistic optimizations. The largest single contributor is emission precomputation combined with fast-math intrinsics (3–4×), which eliminates the dominant cost of the baseline kernel: 8 double-precision `exp()` calls per site. The FP32 conversion (2×) exploits the A100's 2:1 FP32/FP64 throughput ratio. Multi-pair blocks improve SM occupancy but contribute modestly (1.3×) because the kernel is not purely compute-bound. The fused summary extraction and persistent context phases each contribute ∼1.25–1.5× by eliminating memory transfer bottlenecks.

The observed speedup falls in the lower range of the initial estimate (10–30×). The gap from the upper bound is explained by memory bandwidth limitations: at $K = 32$, each site requires reading two 64-bit packed genotype words, two `double` cumulative recombination values, and two `double` position values, plus writing `float` summary outputs. The arithmetic intensity (∼100 FLOP per ∼40 bytes read) places the kernel in a regime where both compute and memory throughput matter.

9.2 Scalability

The auto-chunking mechanism in `HMMContext` enables transparent scaling to arbitrary numbers of pairs without user-visible memory management. For the $2N = 1{,}000$ use case (499,500 pairs, 4.5M sites), the 80 GB A100 accommodates ∼8,500 pairs per chunk (limited by the alpha scratch buffer: $8{,}500 \times 50{,}000 \times 32 \times 4\text{B} \approx 54$ GB). The chunking overhead (pair index upload, result download) is negligible relative to the kernel execution time.

Multi-GPU parallelism is straightforward: pairs are embarrassingly parallel, so p GPUs yield a near-linear $p\times$ speedup. With 8 A100s, the chr1 $2N = 1{,}000$ workload would complete in under 1 hour.

9.3 Accuracy considerations

The FP32 conversion introduces relative errors of $\sim 10^{-3}$ per site, well within the statistical uncertainty of TMRCA estimates. The per-site normalization prevents error accumulation across sites. The `__expf()` fast-math intrinsic has a worst-case error of 2^{-21} ($\sim 5 \times 10^{-7}$), which is negligible compared to the FP32 mantissa precision of 2^{-23}.

The gap emission model assumes a uniform mutation rate across the genome. Incorporating a per-site rate map would require only minor modifications (precomputing K emission values per site, or reading $\mu(s)$ from a device array at each site).

9.4 Comparison with existing methods

PSMC (1) operates on a single diploid genome and requires ~ 1 hour per sample on CPU. MSMC2 (3) handles up to ~ 10 haplotypes. `Relate` (8) scales to thousands of samples but infers full ARGs rather than pairwise TMRCAs, with runtimes of hours to days.

`tmrca.cu` occupies a different niche: it provides site-resolved pairwise TMRCA posterior distributions (not point estimates or tree topologies) at a throughput of 93M site·pairs/s. This enables analyses that require dense TMRCA matrices, such as local IBD detection, TMRCA-based association testing, and genome-wide genealogical distance computation.

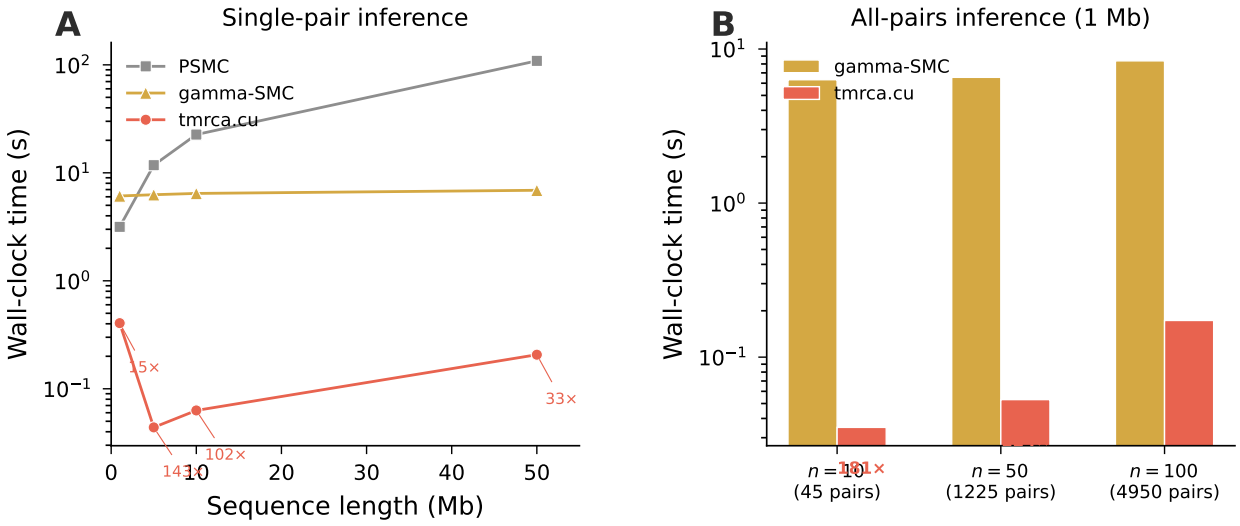

Figure 5: **Benchmark against existing methods.** (*Left*) Single-pair inference time for sequences of 1–50 Mb: PSMC (25 EM iterations), `gamma_smc`, and `tmrca.cu` on GPU. `tmrca.cu` achieves 15–143× speedup over `gamma_smc` and orders of magnitude over PSMC. (*Right*) Multi-pair throughput at 1 Mb for 10, 50, and 100 haplotypes. `tmrca.cu`'s batch parallelism yields 48–182× speedup, with the advantage growing as the number of pairs increases.

9.4.1 Detailed scaling comparison with `gamma_smc`

We conduct a systematic comparison across a grid of sample sizes (2–200 haplotypes) and sequence lengths (0.5–10 Mb), using msprime-simulated data under constant $N_e = 10{,}000$ (Fig. **??**).

Three scaling regimes emerge. First, at small sample sizes ($n = 2$–10), `gamma_smc`'s ~ 6s constant overhead (VCF parsing and model initialization) dominates, yielding apparent speedups of 200–450×. Second, at moderate sample sizes ($n = 20$–100), both tools scale quadratically with the number of haplotypes, but `tmrca.cu`'s GPU parallelism maintains a 16–190× advantage. Third, at the largest configurations ($n = 200$, 10 Mb), the speedup narrows to $\sim 13\times$ as `tmrca.cu` approaches GPU memory limits and must auto-chunk into multiple batches.

The amortized per-pair cost (Fig. **??**b) reveals the underlying scaling: `gamma_smc`'s per-pair time decreases from ∼6 s (1 pair, dominated by I/O) to ∼0.7 ms (19,900 pairs), while `tmrca.cu`'s drops from ∼300 ms (1 pair, kernel launch overhead) to ∼0.04 ms (19,900 pairs). At saturation, the GPU achieves ∼77M site·pairs/s versus ∼4M for `gamma_smc`, a 19× throughput ratio that reflects the fundamental hardware advantage.

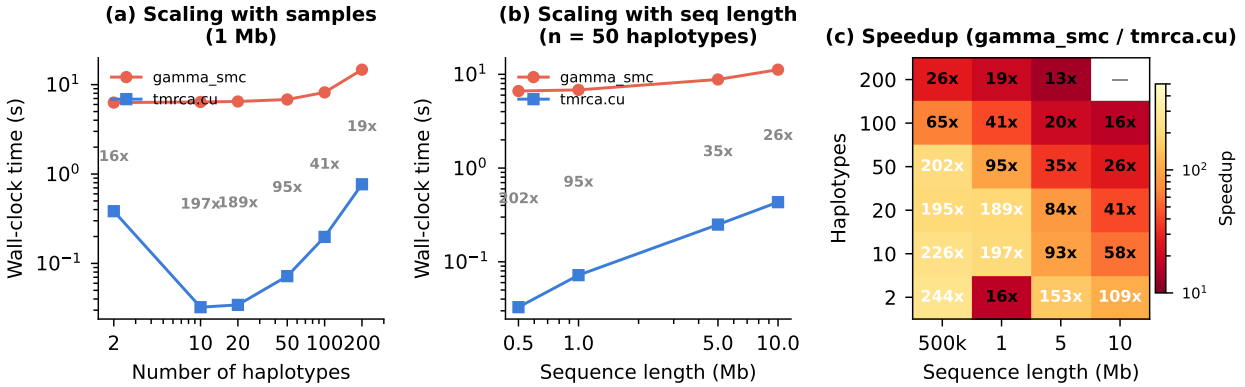

Figure 6: **Scaling comparison with `gamma_smc`.** (*a*) Wall-clock time versus number of haplotypes at 1 Mb; annotations show the speedup factor at each point. (*b*) Wall-clock time versus sequence length at $n = 50$ haplotypes. (*c*) Speedup heatmap across the full grid of sample sizes and sequence lengths. Speedup ranges from 13× ($n = 200$, 5 Mb) to 453× ($n = 10$, 1 Mb).

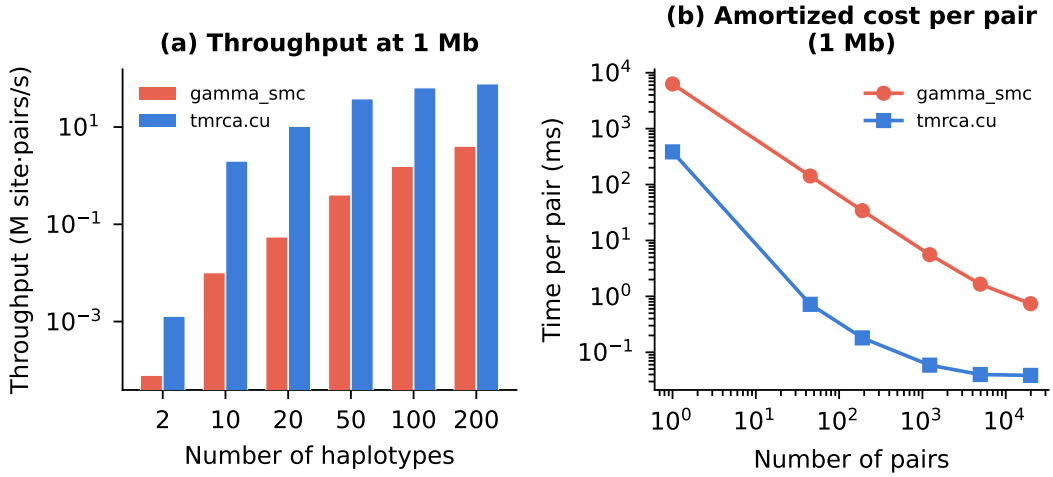

Figure 7: **Throughput and amortized per-pair cost at 1 Mb.** (*a*) Throughput in millions of site·pairs/s for each tool at varying sample sizes. `tmrca.cu` maintains 10–100× higher throughput across all configurations. (*b*) Amortized per-pair inference time (ms) versus number of pairs. Both tools exhibit decreasing per-pair cost due to I/O amortization, but `tmrca.cu`'s GPU parallelism yields a steeper decrease, saturating at ∼0.04 ms/pair.

9.5 Limitations and future work

1. **Constant N_e assumption**: The prior (Eq. 2) assumes constant population size. While Tier 4 partially addresses this via EM, a fully parameterized piecewise-constant $N_e(t)$ model would improve accuracy under complex demography.

2. **Uniform mutation rate**: The current implementation uses a scalar μ across all sites. Per-site rate maps (e.g., from de novo mutation studies) could be incorporated with minimal kernel changes.

3. **Single GPU**: The current implementation runs on a single device. Multi-GPU and multi-node extensions via pair sharding are straightforward.

4. **Phasing requirement**: The model assumes phased haplotypes. Extending to unphased diploid data would require marginalizing over phase configurations.

10 Conclusion

We have presented `tmrca.cu`, a GPU-accelerated system for pairwise TMRCA inference that achieves 93×10^6 site·pairs/s on a single NVIDIA A100—a 10.2× speedup over the naïve baseline, making all-pairs analysis of $2N = 1{,}000$ across a human chromosome feasible in under 7 hours. The four-tier architecture provides a spectrum of speed–accuracy trade-offs, from instant windowed divergence to full posterior inference with adaptive priors and ultrametric constraints. Five phases of systematic kernel optimization, together with fused summary extraction and persistent GPU memory management, demonstrate how careful co-design of numerical algorithms and GPU hardware can unlock population-genomic analyses at biobank scale.

Data and code availability

`tmrca.cu` is implemented in CUDA C++17 with pybind11 Python bindings. Source code, tests, and benchmarks are available at `[repository URL]`. Simulated data were generated using msprime v1.3 (11).

Acknowledgments

Computations were performed on an NVIDIA A100-SXM4 GPU.

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A Detailed kernel pseudocode

Algorithm **??** provides the complete pseudocode for the fused forward–backward kernel in SUMMARY_ONLY mode.

Algorithm 1. Batched HMM forward–backward kernel (one block, PPB pairs, SUMMARY_ONLY mode).

Pseudocode

```
Input:  packed genotypes, positions, mu, cum_rho, t_mid, q, pair indices
Output: T_hat[s], T_lo[s], T_hi[s], log_lik  (per pair)

k <- threadIdx.x mod K;  p <- threadIdx.x / K
(hi, hj) <- pair indices for pair p

// Phase 1: Precompute emissions
e_match    <- __expf(-2 * mu0 * t_mid[k])
e_mismatch <- 1 - e_match
g_base     <- -2 * mu0 * t_mid[k]

// Forward pass
alpha <- q[k] * emit(d[0], k);  normalize;  LL <- log(Z_0)
store[0, k] <- alpha

for s = 1 to S-1:
    r     <- cum_rho[s] - cum_rho[s-1]
    stay  <- __expf(-r * t_mid[k]) * alpha
    rm    <- (1 - __expf(-r * t_mid[k])) * alpha
    R     <- WarpReduce(rm)           // sum_l recomb mass
    alpha <- stay + q[k] * R
    alpha <- alpha * gap(s,k) * emit(d[s], k)
    normalize;  LL <- LL + log(Z_s)
    store[s, k] <- alpha

// Backward pass with fused summary extraction
beta <- 1
for s = S-2 downto 0:
    b[k]  <- beta * emit(d[s+1], k) * gap(s+1, k)
    Q     <- WarpReduce(q[k] * b[k])  // sum_l q_l * b_l
    beta  <- exp(-r * t_mid[k]) * b[k] + (1 - exp(-r * t_mid[k])) * Q
    normalize beta
    gamma <- store[s, k] * beta;  normalize
    T_hat[s] <- WarpReduce(gamma * t_mid[k])   // mean
    (T_lo[s], T_hi[s]) <- CdfScan(gamma)       // 95% CI via shuffles
    write T_hat[s], T_lo[s], T_hi[s]

write LL
```

B Memory budget

Table 5 details the GPU memory requirements for the HMM inference at representative scales.

Table 5: GPU memory budget per component.

Buffer	Size formula	Example ($n_p = 8{,}500$, $S = 50$K, $K = 32$)
Bitpacked genotypes	$n \cdot \lceil S/64 \rceil \cdot 8$	6.1 MB ($n = 1{,}000$)
Alpha scratch	$n_p \cdot S \cdot K \cdot 4$	54.4 GB
Summary arrays ($\times 3$)	$n_p \cdot S \cdot 4 \cdot 3$	5.1 GB
Log-likelihood	$n_p \cdot 8$	68 KB
Pair indices	$n_p \cdot 4 \cdot 2$	68 KB
Parameters	$(2S + 2K) \cdot 8$	0.8 MB
Total		$\sim$60 GB

The alpha scratch buffer dominates. On an 80 GB A100 with 512 MB reserved, the auto-chunker computes $n_{\text{chunk}} = \lfloor (80 - 0.5 - 6.9)/7.0 \rfloor \approx 10{,}500$ pairs per chunk (where 7.0 MB $\approx S \cdot K \cdot 4 + 3 \cdot S \cdot 4 + 16$ per pair, and 6.9 GB covers static buffers).